

Rahul Surve

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OBJECTIVE

Driven by results and deeply motivated to thrive in a globally competitive landscape, I am committed to continuously setting and surpassing ambitious goals. With a keen focus on organizational growth and a dedication to ongoing learning, I aim to contribute meaningfully to the success of the organization while seizing opportunities for professional growth.

PROFESSIONAL SUMMARY

- 5+ years of IT Experience in Data Analysis, Business Analysis, Data Modeling, Data Engineering, Database Management, Business Intelligence and Project Management.
- Experience in Healthcare, Pharmaceutical, Education, Consulting and Sales Industry.
- Extensive experience in Data Management, Data Profiling, Big Data, Market Research, Gap Analysis, Root Cause Analysis, Feasibility Studies, Risks Analysis and Risk management.
- Proficient in analytical programming languages such as SQL, Python, and R through practical application.
- Advanced experience in MS Excel including Macros, Lookups and Pivot Tables and strong experience in data visualization including Power BI, Tableau and IBM Cognos.
- Experience in development methodologies of AGILE and Waterfall.
- Possess strong Project Management skills, keeping in mind the objectives of the project and constraints affecting time, scope, and budget.
- Complete special projects, research requests, or ad hoc tasks as assigned by managers.
- Robust experience in building relationships with diverse clientele, engaging as a specialist between business clients and technical teams and working under pressure with minimum supervision.
- Worked extensively on business development, team management, requirement gathering, converting business requirements to technical requirements and estimation creation with high quality.
- Strong work ethics, taking ownership of all duties and efficient time management skills.
- Demonstrated ability to work both in independent and team-oriented environments with well-developed organizational skills.
- Create and update end-user documentation to support implementation and knowledge transfer.

EDUCATION

Stevens Institute of Technology, Hoboken, USA

Master of Science, Business Intelligence and Analytics

Graduate Certificate, Marketing Analytics

Honors: Beta Gamma Sigma, Provost Masters Fellowship

May 2023

GPA: 3.97

University of Mumbai, Navi Mumbai, India

Bachelor of Engineering, Electronics Engineering

June 2021

CERTIFICATIONS

- Epic Advance Reporting
- Dataiku Core Designer
- Introduction to Python Programming issued by Microsoft Technology Associate
- Data Science and Visualization Foundations Specialization certified by IBM
- Six Sigma White Belt

TECHNICAL SKILLS

- **Programming Languages:** SQL, Python (Pandas, Numpy, SciPy, Sklearn, Matplotlib, PySpark), R
- **Analytical Tools:** Azure, Epic, Dataiku (ETL), Microsoft Excel, JMP, Gephi, Google Analytics
- **Visualization Tools:** Microsoft Power BI, Tableau, IBM Cognos, ThoughtSpot
- **Data Modeling Tools:** Erwin, Alteryx, PowerPivot, Microsoft Visio
- **Database:** Microsoft SQL Server, Oracle (Cerner), MySQL, PostgreSQL
- **Project Management Tools:** Microsoft Office Suite, Jira, Confluence, SharePoint
- **Code Editors:** Jupyter Notebook, Sublime Text, PyCharm, Visual Studio

PROFESSIONAL EXPERIENCE

Virtue Care, Edison, New Jersey Data Analyst (Consulting)

May 2024 - Present

Technical Environment: SQL Server, SQL, Python, Power BI, Excel, Google Forms, Google Sheets, MS Office

Patient Match

- Automated psychologist-patient matching process using Python and SQL Server, reducing manual matching time by 80% and improving overall match performance by 15%.
- Developed and managed SQL Server databases for psychologists and patients, improving data retrieval efficiency by 20% with key data on location, specializations, traits, and availability.
- Engineered a K-Nearest Neighbors (KNN) model for matching patients to psychologists based on traits and comfort levels.
- Streamlined service documentation with automated forms, reducing errors by 15% and ensuring accurate record-keeping for 300+ patient visits.
- Designed Power BI visualizations, including maps and dashboards, to monitor match success rates and patient progress, enhancing decision-making and real-time insights.
- Integrated SQL and Python-based automation, streamlining operations and improving overall process efficiency, leading to a 35% reduction in operational costs.

RWJBarnabas Health, Newark, New Jersey Data Analyst (Financial Services)

June 2023 – May 2024

Technical Environment: Epic Systems (Caboodle, Cadence, Clarity, ASAP), Oracle - Cerner Reporting Bench, SQL, MS Excel, Power BI, MS Visio, PowerPivot, MS Office

Hospital Dashboard

- Developed and refined the Daily Census Dashboard to address irregularities in hospital bed data.
- Segmented all beds according to their respective units, providing real-time overview of bed utilization.
- Optimized code structure for efficiency and ensured inclusion of both occupied and empty beds for comprehensive insights and decreasing coverage issue by 20%.
- Developed VBA macros in Microsoft Excel to automate creation of dynamic hospital dashboard, reducing existing computation time by 90% through automation.
- Successfully transitioned the dashboard to a live environment, overcoming challenges with data visibility and page layout, and provided end-user training to nursing staff.
- Documented development of the dashboard in collaboration with departmental heads, providing detailed explanations for any future changes and help stakeholders understand the process and report generation.

Extended Length of Stay (LOS)

- Analyzed 3 years of electronic health records on Oracle (Cerner database) using SQL to identify patterns linked to extended patient stays, found clear insights, and provided actionable recommendations to management through literature review and benchmarking analysis.
- Utilized Epic's Predictive Discharge Planning Module and Cerner's Predictive analytics solutions to streamline care team focus and anticipate discharge obstacles.
- Employed data visualization tools like Power BI and Power Pivot to present predictive discharge planning insights in a clear and actionable format for non-technical care teams.
- Reduced delays in admission from the Emergency Department, significantly cutting down processing time from 6 hours to 90 minutes.
- Identified critical areas impacting LOS and proposed actionable solutions, including extending lab operation hours, and implementing prioritized queues for discharge-related tests.

Physician Scoreboard

- Calculated the scope of improvement for 250 MDs across 500 diagnoses related groups on EPIC Cogito Analytics and Power BI, utilizing Key Performance Indicators (KPI) to drive resource optimization.
- Utilized physician metrics such as Relative Value Units (RVUs), patient's per day, encounter per session, and others to assess and optimize physician performance.
- Boosted revenue per physician by 5% by utilizing Python with Pandas, SciPy, and NumPy to optimize panel size and patient throughput.

- Implemented regression, clustering, classification, and forecasting, to predict at-risk patients and reduce no-show rates by 12%.
- Identified top 5 physicians based on key performance metrics, enabling recognition of exemplary performance and benchmarking against peers.
- Demonstrated ability to leverage data analytics and performance metrics to drive actionable insights and improve healthcare delivery.

Staffing Dashboard

- Spearheaded the development of a comprehensive staffing optimization project utilizing Power BI for 200 departments encompassing 3000 employees.
- Extracted and integrated data from HR databases, time tracking systems, and budgetary reports into SQL databases, executed data validation, data quality checks, performed missing value imputations, and curated a dataset ensuring data integrity for analysis.
- Analyzed authorized and current working hours, job types, and budgetary constraints to recommend hiring or downsizing decisions.
- Achieved a 5% reduction in total working hours through strategic staffing adjustments based on data-driven insights.
- Reduced overtime expenses by 8% through more efficient allocation of workforce resources.
- Collaborated closely with departmental and non-technical stakeholders to gather requirements and ensure alignment with organizational objectives.

Johnson & Johnson, Titusville, New Jersey **Data Engineer (Advance Analytics)**

May 2022 – August 2022

Technical Environment: Azure, Dataiku, Python, PySpark, Power BI, SQL, Jira, JQL, Confluence, SharePoint, MS Excel, MS Office, Agile

Precision Marketing Project

- Trained an agile and scalable engineering pipeline (ETL) utilizing Python and PySpark in Dataiku to boost Commercial Excellence team's analytical initiatives.
- Proposed and collaborated on business rules with market research team to categorize Health Care Professionals (HCP) into distinct stages depending upon prescribing habits and potential sales revenue.
- Conducted segmentation of 60,000 HCPs into distinct stages based on over 1200 attributes across 6 million rows within the Customer Funnel framework using PySpark and SQL.
- Identified 12,000 HCPs not engaging with patients and 2,400 HCPs prescribing products without patient interactions, indicating potential off-label use or coverage issues.
- Adapted to sudden changes in project direction, transitioning from separate funnels for different drugs to a single dynamic funnel.
- Iterated through 11 versions before finalizing a 6-stage funnel, with considerations for dynamic adjustments.
- Streamlined the funnel to 6 stages and 5 segments from xAI Segmentation, creating micro-segments for optimized marketing strategies.
- Visualized 30 micro-segments for precise marketing strategies in Power BI, enhancing the capacity to target specific groups, resulting in increased engagement and effective marketing campaigns.
- Led a team of 3 members to track HCP movement across stages over time and created opportunity group hypothesis for implementing precision marketing to maximize sales.
- Employed Azure Boards for sprint planning and tracking progress on HCP segmentation and funnel development, ensuring transparency and accountability across the team.
- Leveraged Azure Repos for managing code changes and pull requests, facilitating continuous integration and deployment of Python and PySpark scripts.
- Conducted code reviews and implemented best practices for DevOps, resulting in improved pipeline reliability and reduced deployment time by 30%.
- Enhanced JIRA efficiency by optimizing existing automations, developing new automation rules, and leveraging JQL to conduct complex searches across multiple projects.
- Leveraged Confluence for documentation by creating detailed project documentation, including process flows, technical specifications, and user guides, enhancing team collaboration and knowledge sharing.
- Achieved significant improvements in marketing effectiveness and operational efficiency through innovative data analytics, driving measurable increases in engagement and sales outcomes.

Stevens Institute of Technology, Hoboken, New Jersey
Teaching Assistant (Data Analytics)

August 2021 – December 2022

Technical Environment: Tableau Desktop, Tableau Server, Microsoft Excel, Excel Web, Google Sheets, SharePoint, Agile

- Coached undergraduate students of technology and business majors to practically use analytical and visualization tools of Microsoft Excel and Tableau.
- Collaborated with 4 Teaching Assistants to develop a comprehensive curriculum for 220 students to incorporate knowledge of these tools.
- Instructed students in applying methodologies such as data validation, data importation, process optimization, budgeting, forecasting, ratio analysis, what-if analysis, regression, and various statistical analysis to address real-world challenges in Excel.
- Taught students the application of diverse mapping techniques, charts, plots, joins, tooltips, calculated fields, hierarchies, and dashboards to enhance the clarity of their analysis through visualization in Tableau.
- Developed and implemented advanced analytical projects and case studies using real-world data in Excel and Tableau.
- Conducted regular progress assessments to track student growth and identify areas for improvement, implementing targeted interventions to support individual learning needs.
- Continuously refined teaching strategies to optimize student learning outcomes by integrating new insights from professional development.
- Actively participated in departmental meetings and professional development activities to stay abreast of advancements in analytical tools and pedagogical techniques.

Data Strategix, Navi Mumbai, India
Data Analyst

July 2019 – June 2021

Technical Environment: SQL Server, Tableau, Alteryx, MS Excel, Google Analytics, Google Tag Manager, Web API

- Designed and implemented a robust database structure in SQL Server capable of aggregating vast amounts of data generated by 20+ digital campaigns.
- Utilized Alteryx for data collection and data mapping from multiple sources and employed advanced data manipulation techniques to streamline processes and enhance data integrity.
- Managed, organized and analyzed vast (1M+ rows) daily data from campaigns, to handle data distribution across parameters like gender, age, location, device types, and placements.
- Developed procedures for seamless integration of data provided in Excel format by certain clients, reducing data processing time by 30%.
- Provided the client with 50+ generic SQL queries to extract data for the client, enabling comprehensive data visualization and in-depth analysis in Tableau.
- Visualized key performance indicators (KPIs) such as Click-Through Rate (CTR), Cost Per Click (CPC), Cost Per Acquisition (CPA), Cost Per Mille (CPM), and others, on Tableau, providing valuable insights into campaign performance.
- Assisted with tagging in Google Tag Manager (GTM) by creating and implementing tags, configuring triggers and variables, and ensuring accurate data collection for digital campaign analysis.
- Streamlined data management processes across creative departments, enabling campaign managers to efficiently allocate budgets, leading to a 10% increase in ROI.
- Enabled management to make timely interventions based on accurate reports of campaign performance, resulting in 15% increase in campaign efficiency.
- Analyzed, reviewed, and altered workflows to increase operating efficiency and adapt to new requirements, designed new programs as needed.
- Engineered design solutions and meticulously documented the process to ensure clarity, consistency, and future reference.

Trivia Softwares, Thane, India
Python Intern

April 2020 – June 2020

Technical Environment: Python, Jupyter IDE, Sublime Text IDE, MySQL Server, MS Power BI, Waterfall

- Developed a comprehensive Student Management System using Python, MySQL and Power BI incorporating features for efficient storage, retrieval, and visualization of input data.
- Utilized Python libraries such as Tkinter for GUI development, ensuring user-friendly interaction and seamless navigation within the system.

- Employed MySQL Workbench to import data from Excel files, streamlining data management process.
- Collaborated closely with team members to identify project objectives, translating client needs into actionable development tasks to ensure alignment with project timelines and deliverables.
- Engaged with cross-functional teams to gather requirements, provide technical support, and ensure seamless deployment of the Student Management System within client environments.
- Conducted thorough testing and debugging procedures to validate system functionality and address any potential issues or inconsistencies, ensuring optimal system performance.
- Demonstrated strong problem-solving skills and adaptability in navigating various project complexities, effectively prioritizing tasks and managing competing demands to achieve project objectives.

ATS Learning solutions, Navi Mumbai, India
Summer Intern (IoT - Automation)

June 2019 - July 2019

Technical Environment: Python, C++, Raspberry Pi 3/4, Linux, ThingSpeak, Pushetta, Jupyter IDE, Android API, Arduino Yun

- Spearheaded a dynamic team of 5 members, overseeing the conceptualization, development, and implementation of diverse Home Automation and Internet of Things (IoT) systems.
- Leveraged cloud-based platforms such as ThingSpeak and Pushetta to remotely access and monitor IoT devices, enabling real-time data visualization and analysis from anywhere in the world.
- Orchestrated the deployment and configuration of IoT devices across multiple environments, ensuring interoperability and compatibility with existing infrastructures and systems.
- Collaborated closely with clients to understand their unique requirements and preferences, customizing IoT solutions to meet specific use cases and maximize user satisfaction.